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Zeeshan Khan

Machine Learning Engineer | Computer Vision | MLOps | LLM Applications

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PROFESSIONAL SUMMARY

Machine Learning Engineer with strong experience in computer vision systems, video understanding, and production-scale MLOps pipelines. Specialized in building end-to-end AI systems from data engineering and model training to cloud deployment and monitoring.

Currently contributing to a large-scale robotics video understanding program involving temporal action segmentation and structured video-to-text annotation, with initial production deliverables completed and ongoing expansion discussions for a multi-million-dollar long-term data program.

Extensive experience delivering applied AI solutions in agriculture, surveillance, and industrial domains, including Australia-relevant precision agriculture computer vision systems.
Delivered 170+ freelance ML projects with consistent 5-star ratings across global clients.
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EDUCATION

International Islamic University, Islamabad, Pakistan
Bachelor of Science in Computer Science *Sep. 2018 - June. 2022 | GPA: 3.34 / 4.00*

SKILLS SUMMARY

Languages: Python, SQL, JavaScript, Bash
Frameworks: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Flask, FastAPI
Tools: Docker, Git, ClearML, Excel, Power BI, Tableau, CVAT, Roboflow, Hugging Face, QGIS
Cloud: AWS (Lambda, S3, EC2, ECR, SAM, Elastic Beanstalk), GCP, Azure, Databricks, Snowflake, Banana.dev, Beam, Pipeline.AI
Platforms: VS Code, Cursor, Cluade Code, Jupyter Notebook, Google Colab, Anaconda
Soft Skills: Stakeholder Management, Team Leadership, Communication, Rapport Building

WORK EXPERIENCE

WortelAI
Software Engineer (Machine Learning) *Oct. 2022 - Present*

- Led the development of fully automated object detection and segmentation pipelines for Proofminder, a precision agriculture platform, using OpenMMLab's MMDetection, MMSegmentation, and MMDeploy.
- Built end-to-end MLOps workflows that automated data retrieval from CVAT, dataset splitting, model training, ONNX conversion, and push to Google Cloud Storage; fully integrated with ClearML for experiment tracking and control.
- Designed and deployed microservices for drone image inference and geo-imaging data processing to support real-time agricultural insights.
- Delivered solutions for hundreds of use cases, transforming drone imagery into actionable insights at the plant and leaf level to support sustainable farming practices.
- Built and deployed a serverless image search engine using Qdrant and AWS SAM for scalable visual retrieval.
- Developed a human fall detection and action recognition system using YOLOv5, integrated with real-time CCTV camera feeds for live monitoring.
- Pretrained custom object detection and segmentation models using a large-scale, in-house agricultural dataset containing hundreds of classes (crops, diseased plants, weeds, grasses, etc.), significantly improving downstream performance compared to models pretrained on standard datasets like COCO.
- Worked on fixing the Straighten model for EpicListing *[https://epiclisting.com]* where the Straighten model was inconsistently estimating image rotation — sometimes overcorrecting

already level images, and other times under- or mis-rotating genuinely tilted ones — resulting in visibly misaligned outputs that reduce image quality and client satisfaction. Solved this by training Dino-v3 backbone on a large property dataset.

- Led a data annotation project in collaboration with The Land Institute *[landinstitute.org]*, managing a large agricultural dataset and coordinating annotation workflows for high-quality training data.
- Led a data annotation project in collaboration with The FutureCal Ai, managing a large enginerring drawings dataset and coordinating annotation workflows for high-quality training data.
- Designed and developed interactive dashboards and analytical reports using Power BI and Tableau, enabling real-time insights into model performance, annotation progress, and agricultural data trends for both internal teams and external partners.
- Built an automatic dataset annotation pipeline using the GECO2 model, enabling prompt-based annotation transfer from a single labeled image (COCO format) to large image datasets.
- Developed a batch image segmentation pipeline using the SimpleSeg-Kimi-VL model, enabling prompt-based segmentation and automatic generation of COCO-format annotations.

Turing Enterprises Inc.

LLM Data Scientist (Microsoft|Meta Project) *Oct. 2024 - Apr. 2025*

- Evaluated OpenAI's GPT-4 for data science and machine learning tasks by analyzing its response accuracy and providing ideal benchmark responses to assess performance.
- Conducted comparative analysis of Meta's LLaMA 4 model against other LLMs such as Claude AI, focusing on reasoning quality, accuracy, and usefulness.
- Contributed to the development of Meta's Copilot dataset by designing prompts and authoring high-quality Python and JavaScript code samples to guide model behavior.

Freelance Data Scientist

Fiverr.com [Part-Time] *Jan 2021 – Present*

- Completed over 170 freelance projects in data science, machine learning, and AI with a consistent 5.0-star rating and Level 2 Seller status on Fiverr.
- Delivered end-to-end solutions for diverse client needs, including model development, data preprocessing, computer vision, NLP, and deployment.
- Built a strong reputation for clear communication, fast delivery, and robust, scalable code across a wide range of industry domains.

PROJECT

Missed Tassel Detection in Large Agriculture Fields

- Developed a deep learning pipeline to detect missed female tassels after manual detasseling, reducing manual labor from days to minutes.
- **Tech Stack:** MMDetection, Docker, ClearML, AWS S3

Dropped Fruit Counting for Citrus Greening Detection

- Built an AI model to detect and count prematurely dropped oranges from camera installed in a vehecle, enabling identification green, ripe and rotten oranges.
- **Tech Stack:** Object Detection, Deep Learning, ClearML, AWS, MMDetection.

Chilean Needle Grass Identification in Mixed Vegetation in Australia

- Designed a challenging image segmentation solution to detect Chilean Needle Grass among similar crops and weeds ("green-on-green" problem).
- Worked with domain experts to develop discriminative features and improve model precision.
- **Tech Stack:** Image Segmentation, Python, Docker, ClearML, AWS.

Human Fall Detection from CCTV Footage

- Created a real-time fall detection system using YOLOv5 and custom post-processing, deployed on live CCTV camera feeds in warehouses and public areas.
- Improved worker and public safety by enabling automated alerts from live surveillance.

Plant Counting in Drone Videos using RFDet & Roboflow Tracker

- Fine-tuned the RFDet object detection model on high-resolution drone imagery to accurately

count plants in agricultural fields.
- Integrated Roboflow’s object tracking repository to track and de-duplicate detections across frames, enabling reliable plant counts even with drone motion.
- Designed a custom post-processing pipeline to filter overlapping detections and report accurate field-level statistics.
- **Tech Stack:** RFDet, Roboflow Tracker, Drone Imagery, PyTorch, Deep Learning.

Custom Object Detection Pipeline with DFine Transformer
- Fine-tuned the DFine object detection model on a domain-specific dataset by adapting data from CVAT’s COCO format to DFine’s required input format using a custom conversion script.
- Streamlined data preprocessing, annotation conversion, and model training to build a robust object detector for complex real-world imagery.
- Evaluated and validated the model’s performance using task-specific metrics and visual diagnostics.
- **Tech Stack:** DFine (Hugging Face Transformers), CVAT, COCO Format, Python, PyTorch.

Model Deployments
- Deployed machine learning models using Python frameworks (PyTorch, TensorFlow, ONNX) to AWS Serverless Application Model (SAM) for scalable, cost-efficient model serving.Additionally, implemented machine learning model deployment using C# with ONNX to integrate models into enterprise-level applications.
- Deployed GPU-based ML models (Torch, TensorFlow, ONNX) to serverless platforms like Banana.dev, Beam, and Pipeline.AI, enabling low-latency, high-throughput inference in production environments.

Robotics Video Understanding Program (Pre-Contract / Expansion Stage)
- Delivered initial production-grade outputs for a robotics lab video understanding system involving action segmentation and structured annotation
- Participating in advanced-stage discussions for a proposed large-scale video annotation program (~300,000–1,000,000 hours of video data; estimated $3M–$10M USD budget, scalable upward)
- Designing system architecture for:
  - temporal action segmentation in video streams
  - structured natural language descriptions of human actions (household tasks)
  - scalable annotation pipelines for high-volume datasets
- Contributing to planning for workforce expansion (estimated 500–1000 annotators) and annotation quality assurance systems
- Involved in early-stage discussions for potential company formation with partner organization(s)

Algorithmic Contribution
- Developed geometric clustering algorithm for agricultural tassel classification.
- Handles irregular spatial distributions using angular density estimation and directional grouping.

COURSES & CERTIFICATIONS

- Machine Learning Specialization – Coursera (by Andrew Ng)
- Deep Learning Specialization – Coursera (by Andrew Ng)
- Practical Deep Learning for Coders – fast.ai (by Jeremy Howard)
- CS231n – Convolutional Neural Networks for Visual Recognition – Stanford Online (by Andrej Karpathy)
- TensorFlow 2 for Deep Learning Specialization – Coursera (by Dr. Kevin Webster)
- PyTorch Specialization, Python for Data Science, MySQL for Data Science – YouTube
- Power BI: From Zero to Hero, Tableau for Data Visualization – YouTube